$$\min_{\mathbf{x}\in\mathcal{X}} \; c^\top \mathbf{x} \;+\; \max_{\mathbf{w}\in\mathcal{W}(\Gamma)} \; \min_{\mathbf{y}\in\mathcal{Y}(\mathbf{x},\mathbf{w})} \; d^\top \mathbf{y}$$

1 Invest
here-and-now: generation and storage capacity

2 Transmission blackout
adversary: worst outage window

3 Dispatch
wait-and-see: generation, storage, load shedding

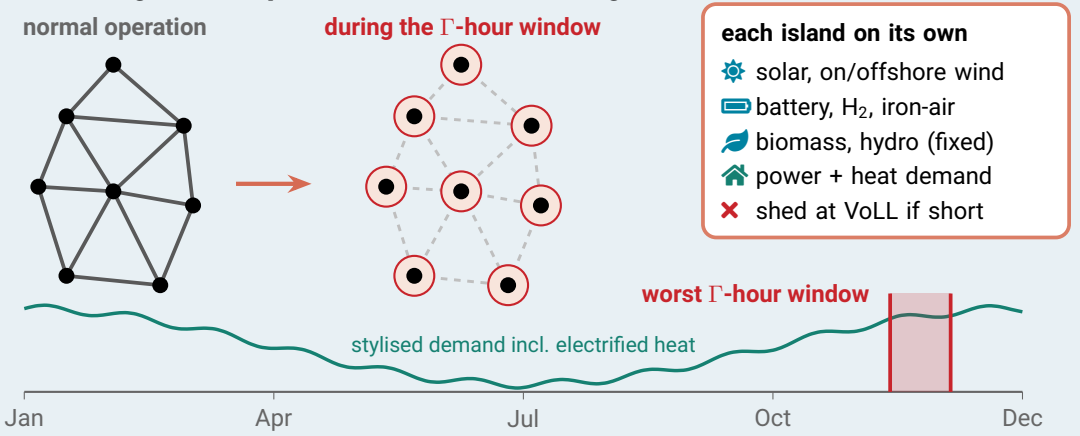
Uncertainty set

$\mathcal{W}(\Gamma) = \{\mathbf{w} \in \{0,1\}^T : \; w_t = 1 \text{ on one contiguous block of } \Gamma \text{ hours}\}$

$w_t = 1 \;\Rightarrow$ **all** AC lines and DC links are out: every region becomes an island and must ride through on local generation and storage.

Capacity is committed before the timing is known; dispatch adapts once it is. No probabilities needed: the worst window is found endogenously.

Modelling the temporal transmission outage



each island on its own
- solar, on/offshore wind
- battery, H_2, iron-air
- biomass, hydro (fixed)
- power + heat demand
- shed at VoLL if short

Solving the ARO problem via Column-and-Constraint Generation (C&CG)

Master (LP)
$\min_{\mathbf{x}} \; c^\top \mathbf{x} + \max_{k\le K} \text{opex}_k(\mathbf{x})$
one dispatch per window found so far; CVaR with ω=1 makes the max exact/robust
$\Rightarrow$ **lower bound**

design $\mathbf{x}^*$ →

← worst window $\mathbf{w}^*$

Subproblem (worst window)
$\max_{\mathbf{w}\in\mathcal{W}(\Gamma)} \min_{\mathbf{y}} \; d^\top \mathbf{y}$ given $\mathbf{x}^*$
exact enumeration: one LP per start t_0, in parallel (alternatively, a dualised single MILP)
$\Rightarrow$ **upper bound**

repeat until $(\text{UB} - \text{LB})/\text{UB} < \epsilon = 10^{-4}$

Case study (Germany)

Network
PyPSA-Eur Germany, 8 (arbitrary) nodes, 2050

Time
full year, 3-hourly, weather year 2013

Heat
electrified: heat pumps + resistive, exogenous load

Invest
solar, wind, battery, H_2, iron-air; + methanol reserve (scenario)

Policy
zero CO_2: no fossil backup, no imports from neighbours

Sweep
$\Gamma \in \{6, 24, 72, 168, 336\}$ h
VoLL 10 000 €/MWh